

Luxury E-Commerce Digital Ecosystem Transformation

Shopify Plus | Product Operations | AI-Assisted Content | Performance Optimization

ROLE: PRODUCT OWNER / DIGITAL & AI SYSTEMS

2021-2025

The engagement

A luxury multi-brand retailer faced a 77% marketing budget reduction, a sharp decline in Google Search revenue, falling social engagement, failing Core Web Vitals, and a product-content operation that could not scale. The mandate was not to spend more. It was to identify the structural failures, rebuild the operating foundation, and document a system the business could run under real constraint.

77%

MARKETING BUDGET REDUCTION

Growth had to come from stronger systems, not higher spend.

RED

CORE WEB VITALS STATUS

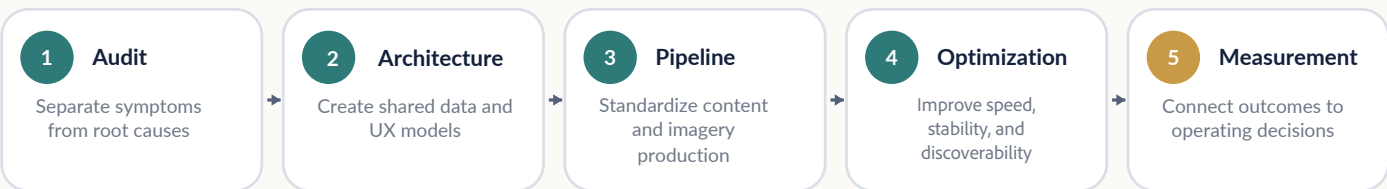
Performance and layout stability were failing before optimization.

1-OFF

PRODUCT WORKFLOWS

PDPs, imagery, metadata, and uploads required repeated manual decisions.

What I rebuilt



CONFIDENTIALITY NOTE

All diagrams, interfaces, product examples, and workflow visuals in this case study were recreated and anonymized. No client admin views, customer data, proprietary spreadsheets, logos, or third-party product photography are shown.

From fragmented tasks to an operating system

The work connected product data, content, imagery, performance, and measurement into one repeatable production model.

BEFORE

Disconnected execution

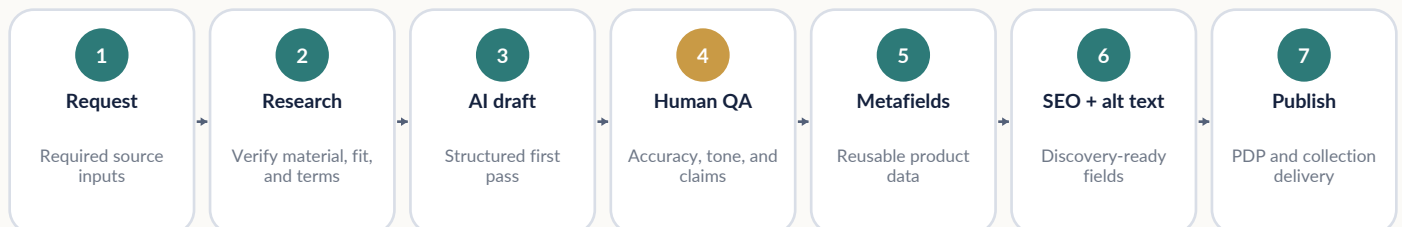
- ✗ PDP content varied by product and vendor source
- ✗ Product uploads depended on manual writing and one-off cleanup
- ✗ No shared metafield architecture or taxonomy
- ✗ Image edits were handled product by product
- ✗ Performance issues were treated separately from asset workflow
- ✗ No documented QA or handoff standard

AFTER

Structured production model

- ✓ Standardized PDP hierarchy and reusable content requirements
- ✓ Metafields captured fit, fabric, silhouette, color, occasion, and trend
- ✓ AI-assisted drafting paired with mandatory human review
- ✓ Documented ghost-image protocol and file-delivery standards
- ✓ SEO metadata, alt text, collections, and product records shared one model
- ✓ Measurement established before optimization decisions

Repeatable product-content pipeline



System-level design principle

The same product facts should not be rewritten independently for every channel. A single governed data model can support the PDP, collection logic, SEO metadata, alt text, upload files, campaign language, and future automation.

PDP

Collections

Search

Alt text

Email

Social

Product feeds

Reporting

Recreated product page before and after

The example below uses fictional product data. It demonstrates the information architecture and content standards without exposing a client page or designer imagery.

BEFORE

PRODUCT

Ivory Blouse

Beautiful blouse with long sleeves and tie detail. A great piece for any wardrobe.

DETAILS

Imported. Dry clean. True to size.

Description

Shipping

More info



AFTER

FICTIONAL LABEL

Fluid Tie-Neck Blouse - Ivory

A softly draped blouse with a relaxed silhouette, gathered sleeves, and a removable neck tie.

PRODUCT DETAILS

MATERIAL

Silk-viscose blend

FIT

Relaxed; fluid through body

SILHOUETTE

Long sleeve; gathered cuff

DETAILS

Removable tie; concealed placket

CARE

Professional clean

DISCOVERY LAYER

SEO title, meta description, collection assignment, and image alt text all use the same verified product facts.



What changed

NORMALIZED IDENTITY

Consistent product naming and designer/category logic.

SCANNABLE HIERARCHY

Summary first, then fit, fabric, silhouette, details, and care.

REUSABLE FACTS

The same verified data populated metafields, collections, SEO, and alt text.

CONTROLLED LANGUAGE

Luxury tone without vague filler or unsupported product claims.

Portfolio case study - visuals recreated and anonymized to protect confidential information and third-party IP.

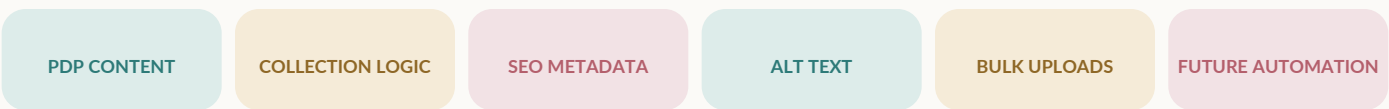
3

A shared metafield architecture for scale

The model turned scattered product facts into reusable structured data. The visual uses fictional values and generic fields.



The same data powered multiple outputs



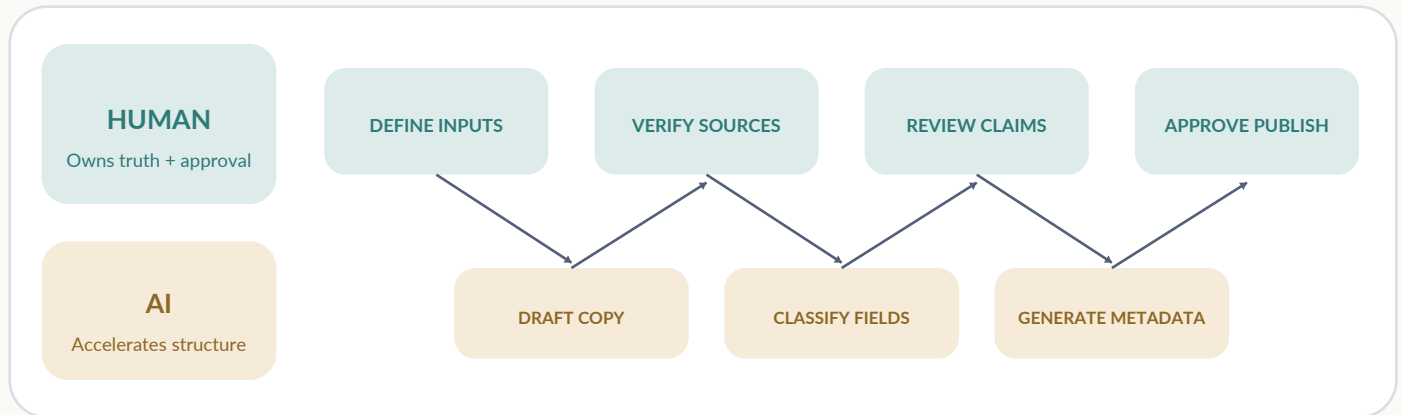
Anonymized upload example

PRODUCT	FIT	FABRIC	COLOR	COLLECTION
Fluid Tie-Neck Blouse	Relaxed	Silk blend	Ivory	Modern Workwear
Soft Tailored Trouser	Straight	Wool blend	Charcoal	Wardrobe Foundations
Bias Midi Skirt	Skimming	Viscose satin	Cocoa	Tonal Dressing

Design outcome: the catalog no longer required a separate content decision for every product and every channel. Structure created the foundation for scale, consistency, and later automation.

Automation with human control

AI accelerated the first draft and classification work. Human review remained mandatory for product accuracy, luxury tone, and publication readiness.



Fictional content example

UNSTRUCTURED INPUT

"Ivory blouse. Silk blend. Tie neck. Long sleeve. Dry clean."

RISKS

- Generic copy
- No fit context
- No reusable structure
- Weak search language

GOVERNED OUTPUT

A fluid ivory blouse with a relaxed silhouette, gathered sleeves, and a removable neck tie. The silk-viscose blend creates a soft drape suited to modern workwear and event dressing.

REUSED AS

- PDP summary
- Fit and fabric fields
- SEO metadata
- Image alt text

Publication governance

- ✓ Material, construction, fit, and care verified against source information
- ✓ No unsupported performance, sustainability, or origin claims
- ✓ Designer terminology and luxury tone preserved
- ✓ Metadata and alt text describe the actual product rather than keyword stuffing
- ✓ Human reviewer owns the final decision and correction history

A documented ghost-image production system

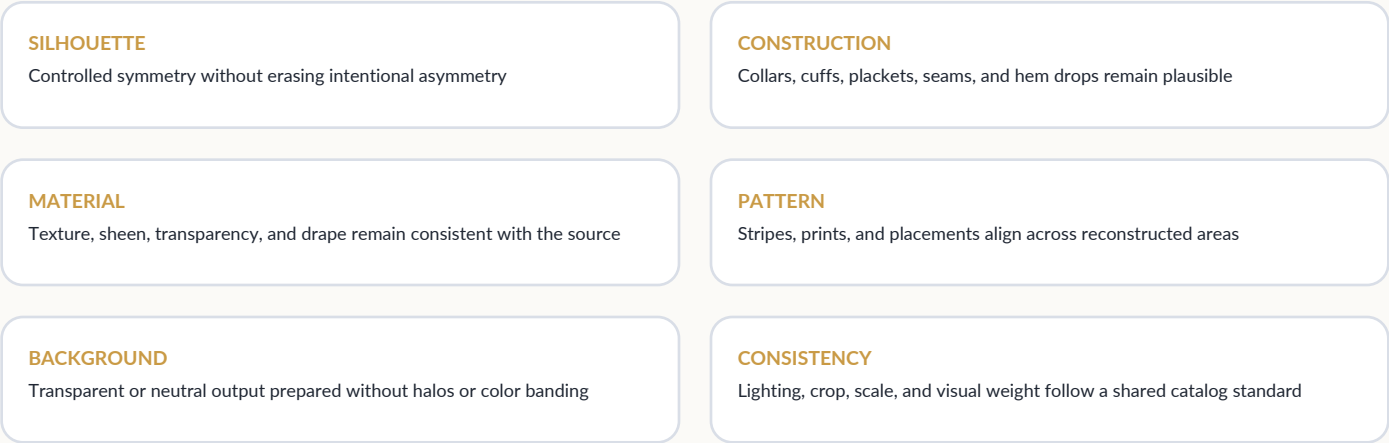
No designer or client imagery is shown. The garment illustrations are original placeholders used to demonstrate the workflow and QA standard.



Documented protocol



Quality standards



Performance and business outcomes

Metrics below are recreated from contemporaneous engagement reporting. The visual does not reproduce private dashboards or account-level data.

CORE WEB VITALS

Site status moved from failing to passing after launch.

RED



GREEN

Documented technical improvement

LCP

28%

faster



INP

~70%

faster



CLS

90%

more stable



Documented outcomes during tenure

700 -> 10,000

ACTIVE CONSENTED SUBSCRIBERS

Approx. +1,329% growth during the engagement.

+550%

CAMPAIGN-ATTRIBUTED CONVERSIONS

Year-one growth; followed by +98% in year two and +60% in year three.

+209%

GROSS MARGIN

Reported improvement while the business converted fewer, higher-quality transactions.

-77%

MARKETING BUDGET

The operating system was built under material resource constraint.

Interpretation: these are engagement-level outcomes observed during the same tenure, not a claim that one isolated intervention caused every result. The case study focuses on the systems built, the operating decisions enabled, and the measurable changes recorded.

Implemented work vs. roadmap

This distinction protects the credibility of the case study and reflects only work completed or documented before the role ended.

IMPLEMENTED / OPERATIONALIZED

Production systems

- ✓ Cross-channel diagnostic baseline and measurement
- ✓ PDP information architecture and content standards
- ✓ Metafield schema and reusable product taxonomy
- ✓ Structured product-upload and bulk-upload workflows
- ✓ AI-assisted PDP, metadata, classification, and alt-text process
- ✓ Human review and publication governance
- ✓ Ghost-image production protocol and QA standards
- ✓ Asset and template optimization supporting Core Web Vitals
- ✓ Lifecycle, subscriber, attribution, and profitability analysis
- ✓ Documentation for repeatability and team handoff

PLANNED / PARTIAL / NOT CONFIRMED

Next maturity stage

- + Person and ImageObject schema for staff E-E-A-T
- + Broader schema and breadcrumb rollout across page types
- + Expanded FAQ content and homepage editorial module
- + Formal entity mapping across brand, staff, designers, and content
- + Editorial clusters with Article and HowTo schema
- + Full cross-channel dashboard and reporting cadence adoption
- + Automated metafield population and visual-recognition tagging
- + PDP auto-generation with variant logic
- + AI-powered trend-to-metadata mapping

Roadmap status by phase

FOUNDATION

Substantially implemented

TECHNICAL + PDP

Mostly implemented

E-E-A-T + EDITORIAL

Strategy / partial

LONG-TERM ECOSYSTEM

Strategy / partial

Portfolio language should match the evidence: “implemented,” “built,” “operationalized,” or “recommended” are used deliberately rather than interchangeably.

Systems thinking that moves from diagnosis to delivery

The engagement demonstrates the ability to distinguish symptoms from root causes, design a shared operating architecture, implement it in a live environment under constraint, and connect the work to measurable outcomes.

01 SYSTEMS DIAGNOSIS

Separated traffic, feed, UX, performance, retention, and profitability problems instead of treating decline as one issue.

02 PRODUCT OWNERSHIP

Defined requirements, information architecture, workflows, QA, handoffs, and operating standards across teams.

03 DATA ARCHITECTURE

Created reusable product structure that connected PDPs, collections, metadata, alt text, and uploads.

04 AI WORKFLOW DESIGN

Used AI where it accelerated production while preserving human control over truth, tone, and publication.

05 MEASURABLE OPTIMIZATION

Improved site performance, production consistency, audience growth, and decision quality under reduced spend.